

Presented by Bitergia & Toscalix Consulting

AGL's Delivery Process Performance: A Data-Supported View

AGL AMM Summer 2025, Berlin

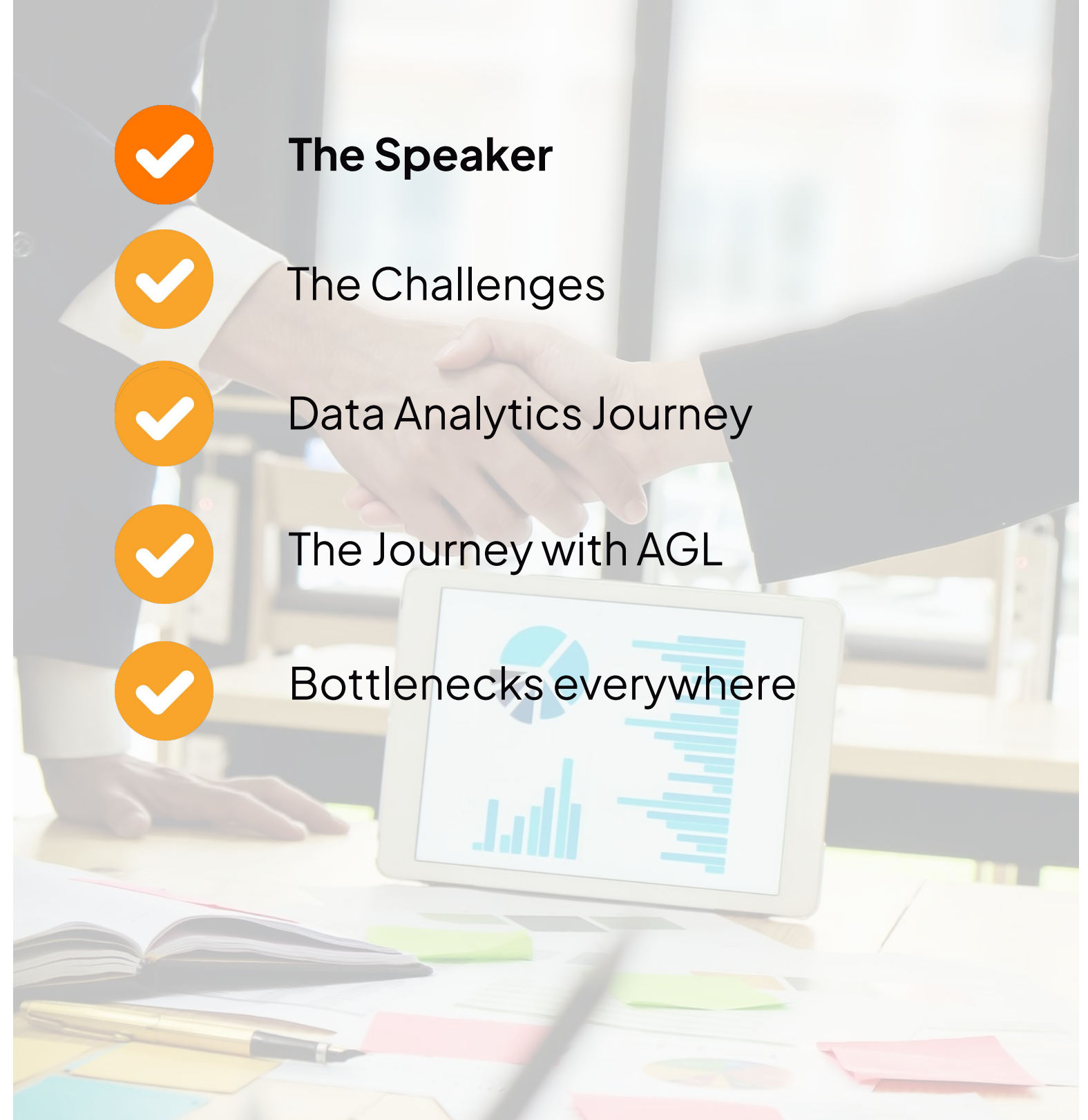
Software Analytics for Delivery Performance Excellence

Luis Cañas-Díaz
Data Consultant at Bitergia
lcanas@bitergia.com
AGL AMM 2025. Berlin
2025-07-10

The Speaker



- ✓ The Speaker
- ✓ The Challenges
- ✓ Data Analytics Journey
- ✓ The Journey with AGL
- ✓ Bottlenecks everywhere





@canasdiaz

Luis Cañas-Díaz

- Empirical Software Engineering background
- GrimoireLab/CHAOSS Contributor
- Co-Founder and Lead Data Consultant at [Bitergia](#), a software development analytics company
 - Core competencies on producing high quality data, incremental support, historical background. Contextualizing data into different verticals
 - InnerSource and open source consulting services and improving the dynamics of collaboration across units. Bringing visibility to Upper Management on development activities



<https://www.linkedin.com/in/canasdiaz/>



<https://github.com/canasdiaz>



Highly-regulated verticals
experienced
engagements

Health
Banking/Finances
Automotive



WHY BITERGIA?

- +15 years of experience with software dev. analytics
- Strong focus on open source and InnerSource: Co-founder of the CHAOSS project and partner of the InnerSource Commons Foundation
- Full service for 100% open source metrics solution, OSS supply chain risk model and GrimoireLab main maintainer
- Fully GDPR compliant
- Flexible team and engagement options
- ISO certified for 9001 and 27001

Contact the Delivery Performance Analytics team:



delivery-performance@bitergia.com



[Delivery Performance Analytics - Bitergia website](#)

THE EXPERTS

BITERGIA



Seasoned in **software development**
at scale analytics

Rooted in **open source** and **InnerSource** community
and ecosystem management analytics

Leaders shaping standard **open-source tools** for
advanced data analytics in diverse industries

TOSCALIX

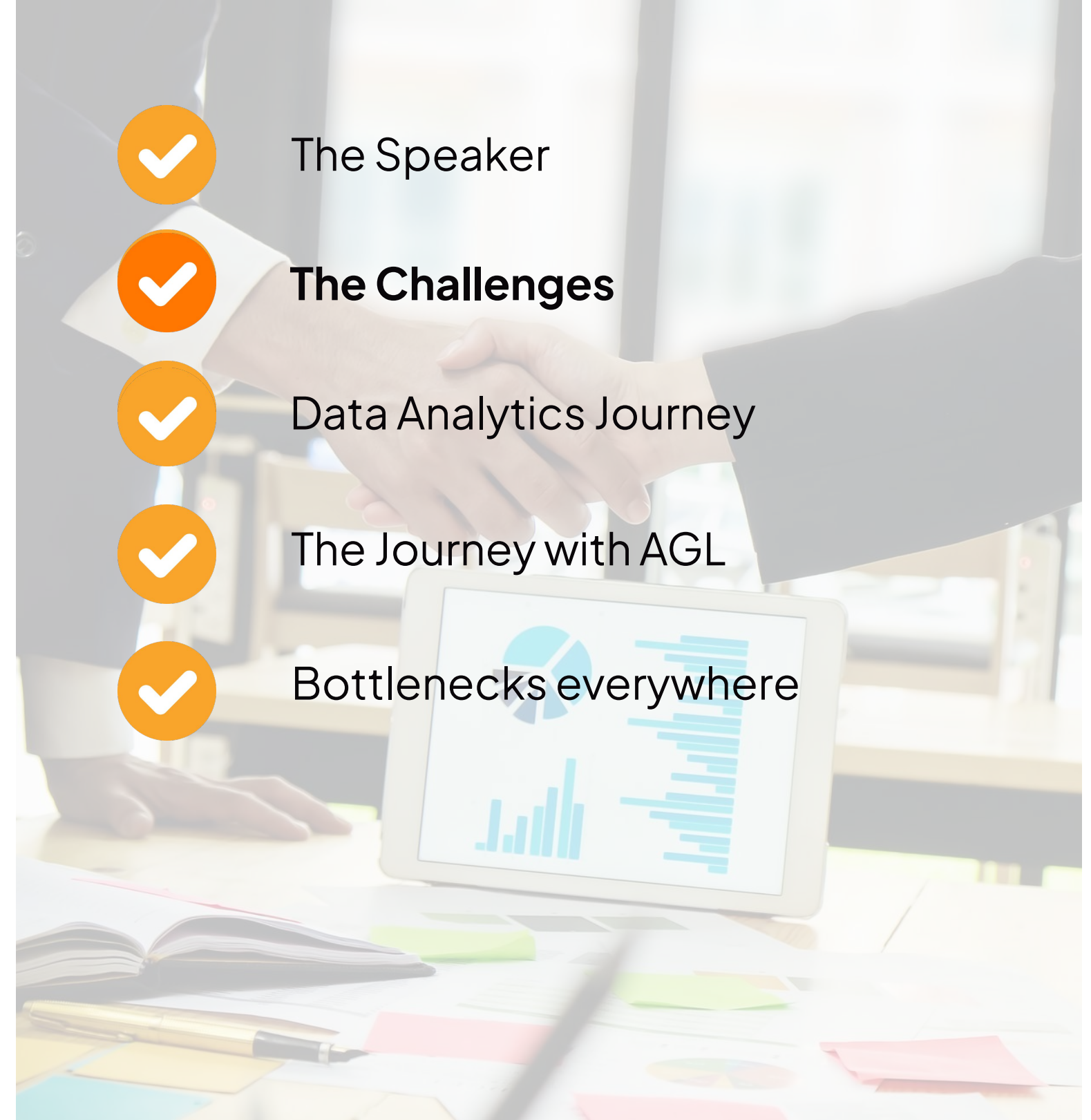


Software defined products **delivery**
performance expert

Experience in **several industries**:
embedded, automotive, industrial, IoT, enterprise...

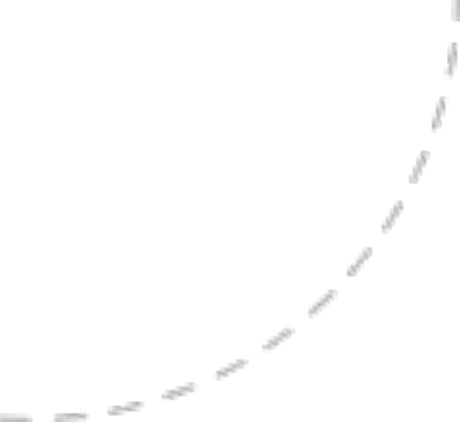
Experience as **manager and consultant** in
several first class organizations

Some context first: The Challenges



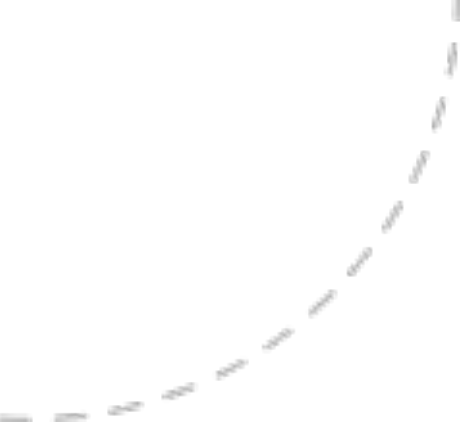
The Challenges

- 01** Quality Software
- 02** Reduce time-to-market
- 03** Frustration / workforce well-being
- 04** Bringing visibility to Upper Management on the software production chain



To face these challenges, organizations must leverage multiple factors:
automation, agile principles, improved hardware testing, and advanced data
analytics — applied throughout the entire production process.





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Common recommendations

01 Be careful with the bottom-up approach

02 Be careful using metrics as a service

03 Bring strategy to the data thinking

04 Metrics is not the discussion, that's a red flag

05 Business Intelligence applied to software development

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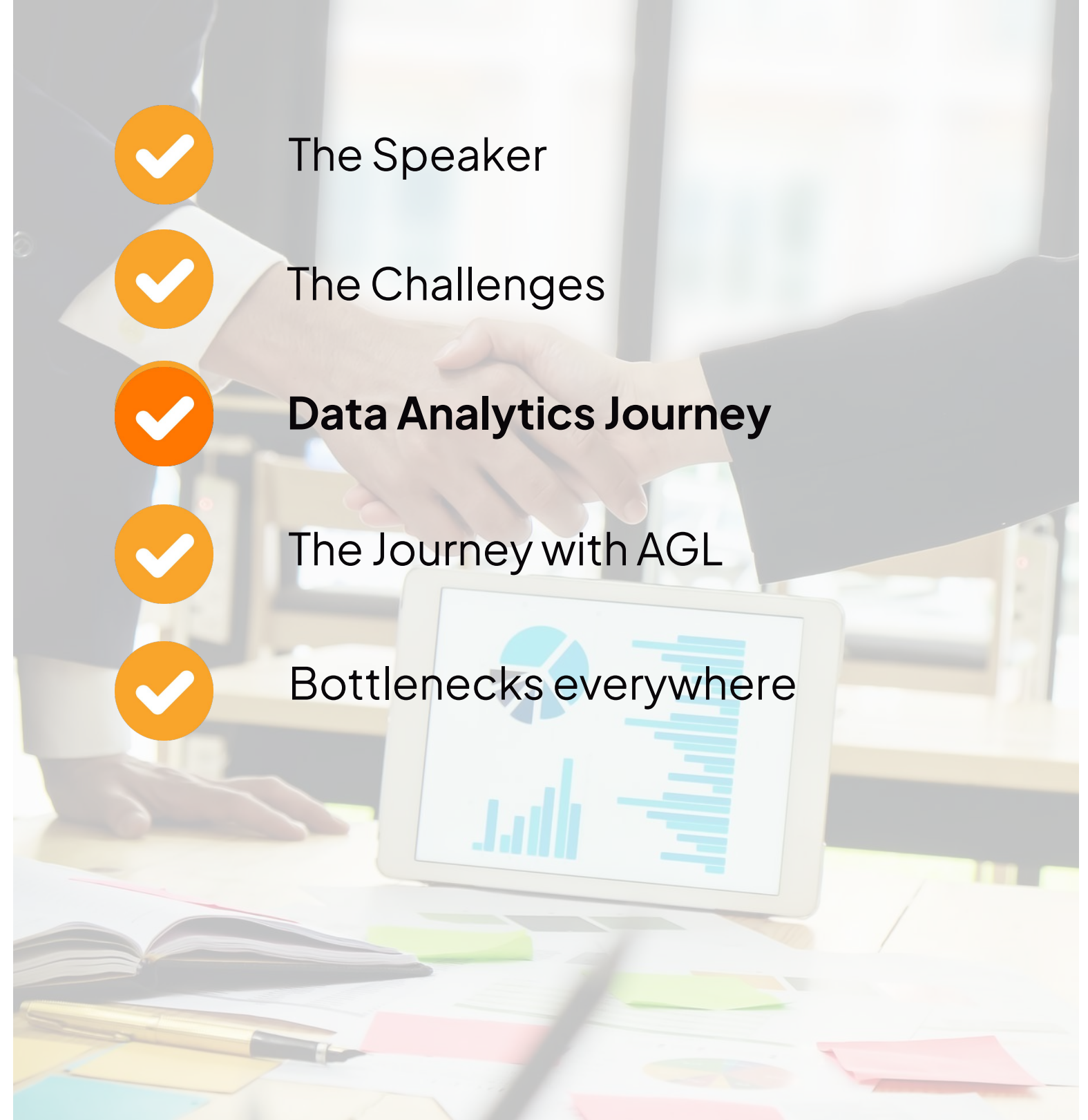
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Data Analytics Journey



- ✓ The Speaker
- ✓ The Challenges
- ✓ **Data Analytics Journey**
- ✓ The Journey with AGL
- ✓ Bottlenecks everywhere



BI IS A JOURNEY

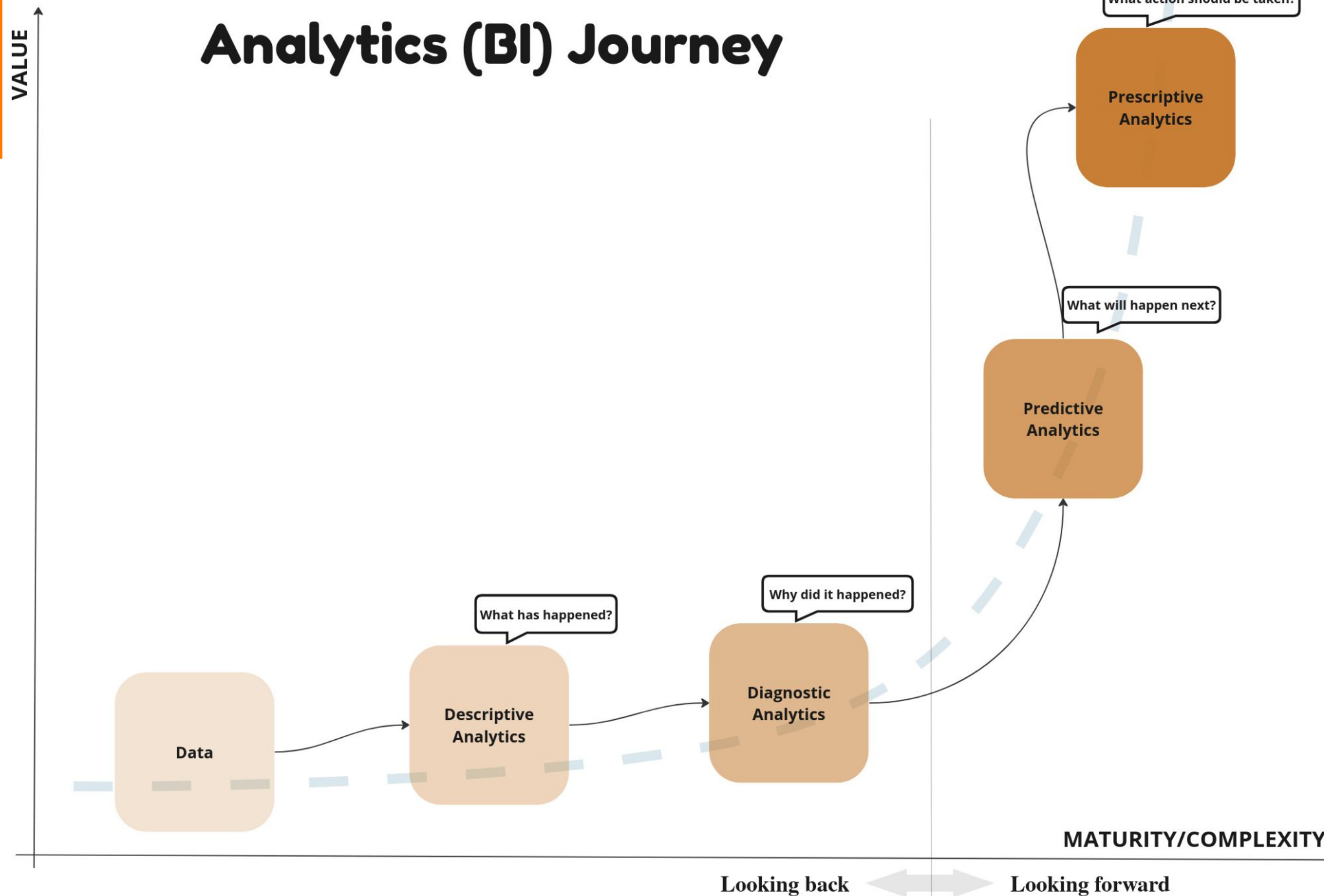
The BI Journey enables organizations to move from reactive problem-solving to proactive, strategic execution, ensuring faster, more reliable, and cost-effective production of software, in four steps

1. **Data Foundation:** collect and transform raw software production data into consumable data, accessible across your organization. Unlock visibility into your development ecosystem.
2. **Understanding the Present - From What to Why:** provide a clear picture of your current software production landscape. revealing trends (**descriptive** analytics) and uncovering the root causes of performance inefficiencies (**diagnostic** analytics).
3. **Anticipate the future - Forecast:** move beyond reactive analysis to predictive modeling (**predictive** analytics) to anticipate delivery risks before they materialize.
4. **Driving Optimized Outcomes:** provide data-driven recommendations to optimize decision-making (**prescriptive** analytics) to improve both delivery and organizational performance.



BI Journey

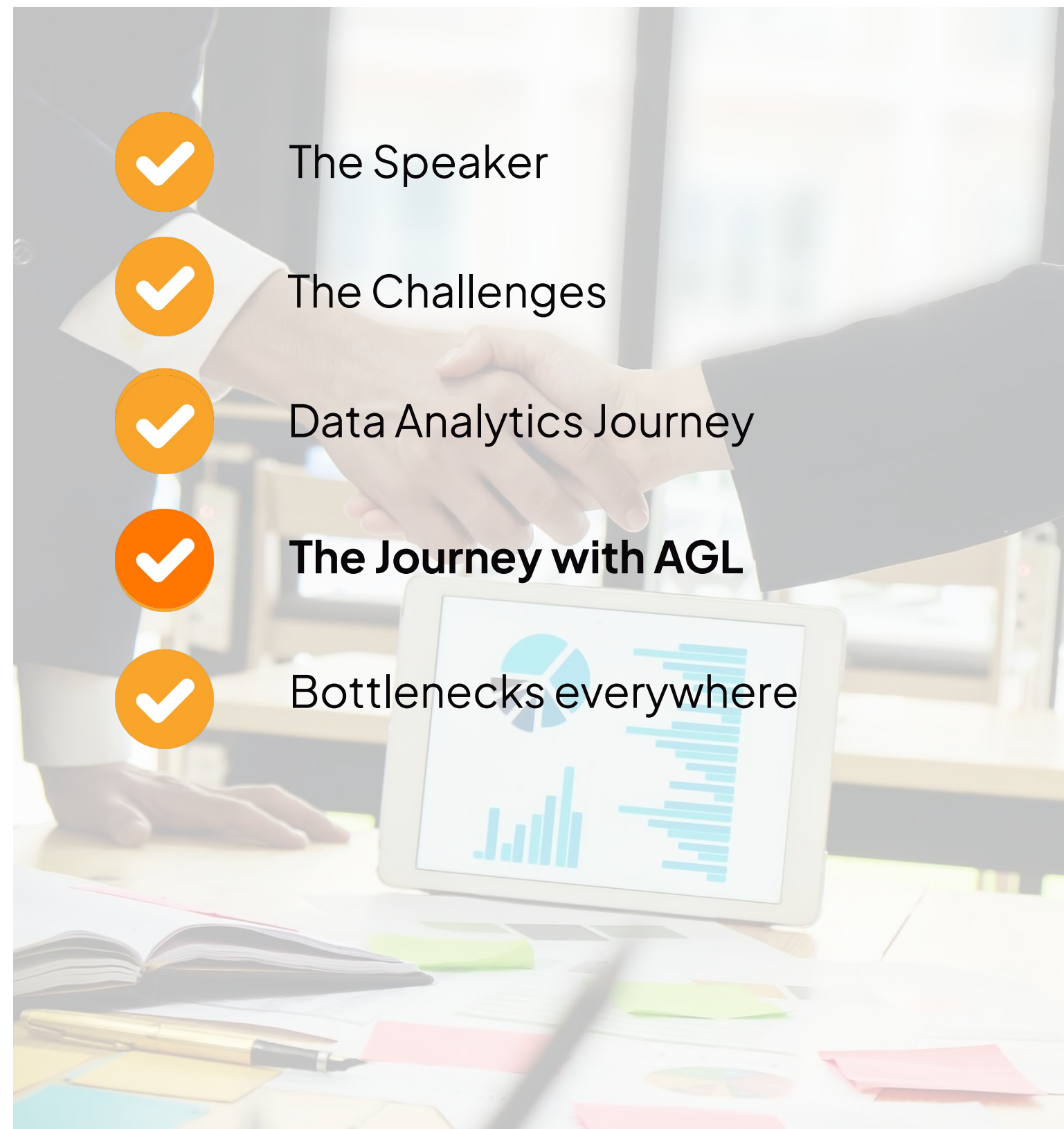
Analytics (BI) Journey

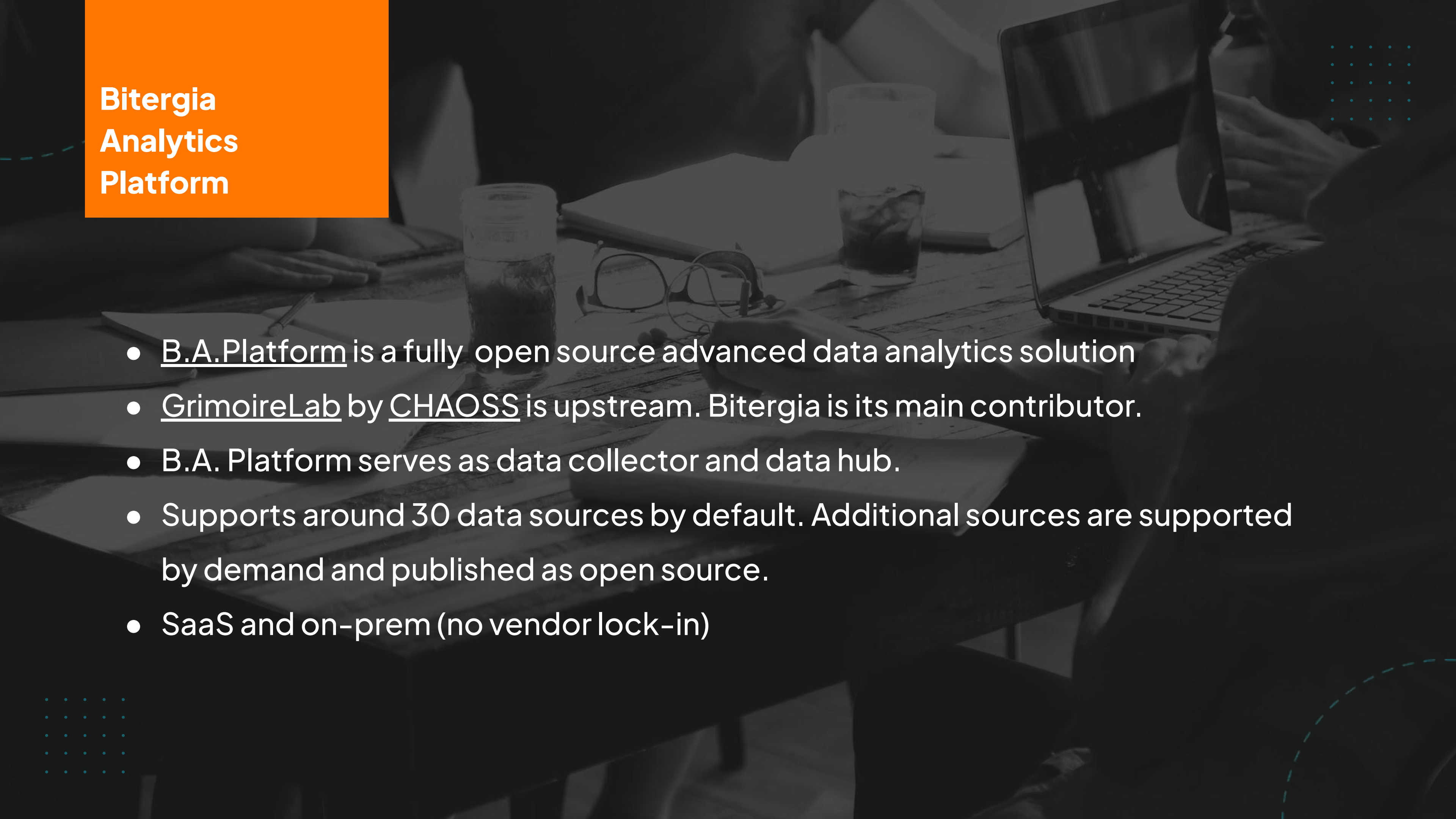


The Journey with AGL



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Bitergia Analytics Platform

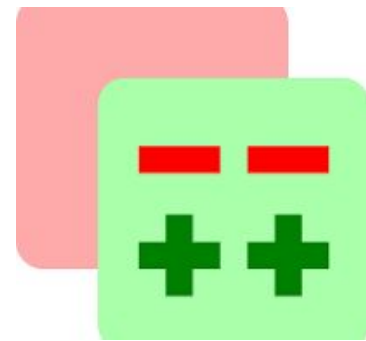
- B.A.Platform is a fully open source advanced data analytics solution
- GrimoireLab by CHAOSS is upstream. Bitergia is its main contributor.
- B.A. Platform serves as data collector and data hub.
- Supports around 30 data sources by default. Additional sources are supported by demand and published as open source.
- SaaS and on-prem (no vendor lock-in)

01

Data



Git commit logs



Gerrit changesets + pathsets



Jenkins

Jenkins jobs + builds

Descriptive Analysis: Delivery Process

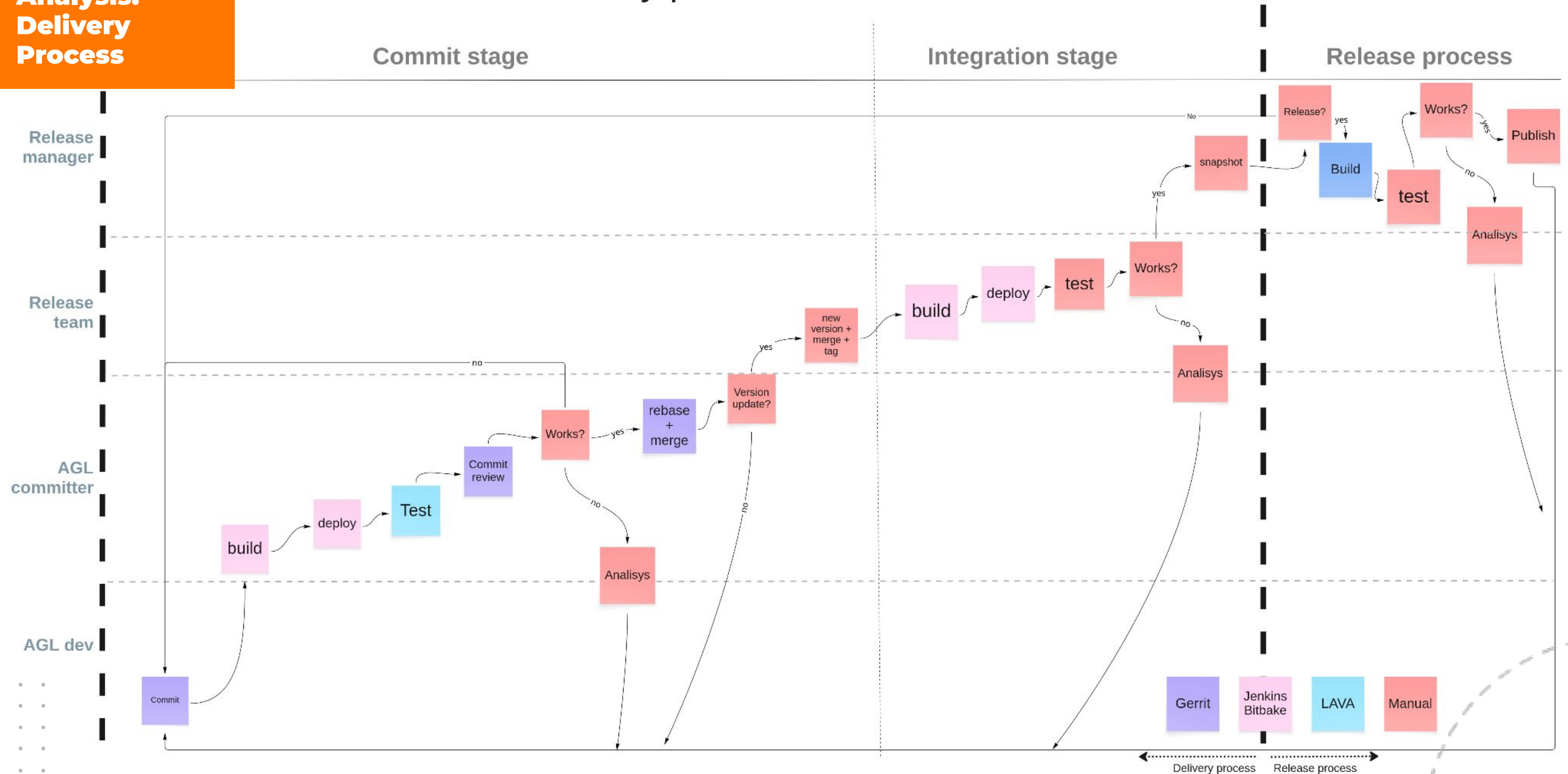
Delivery process



02

Descriptive Analysis: Delivery Process

AGL delivery process model. Iteration 2





The metrics.

Throughput Measurements:

- Lead time: time your code needs to reach the end of the process
- Time Interval: time between two successful executions

Stability Measurements:

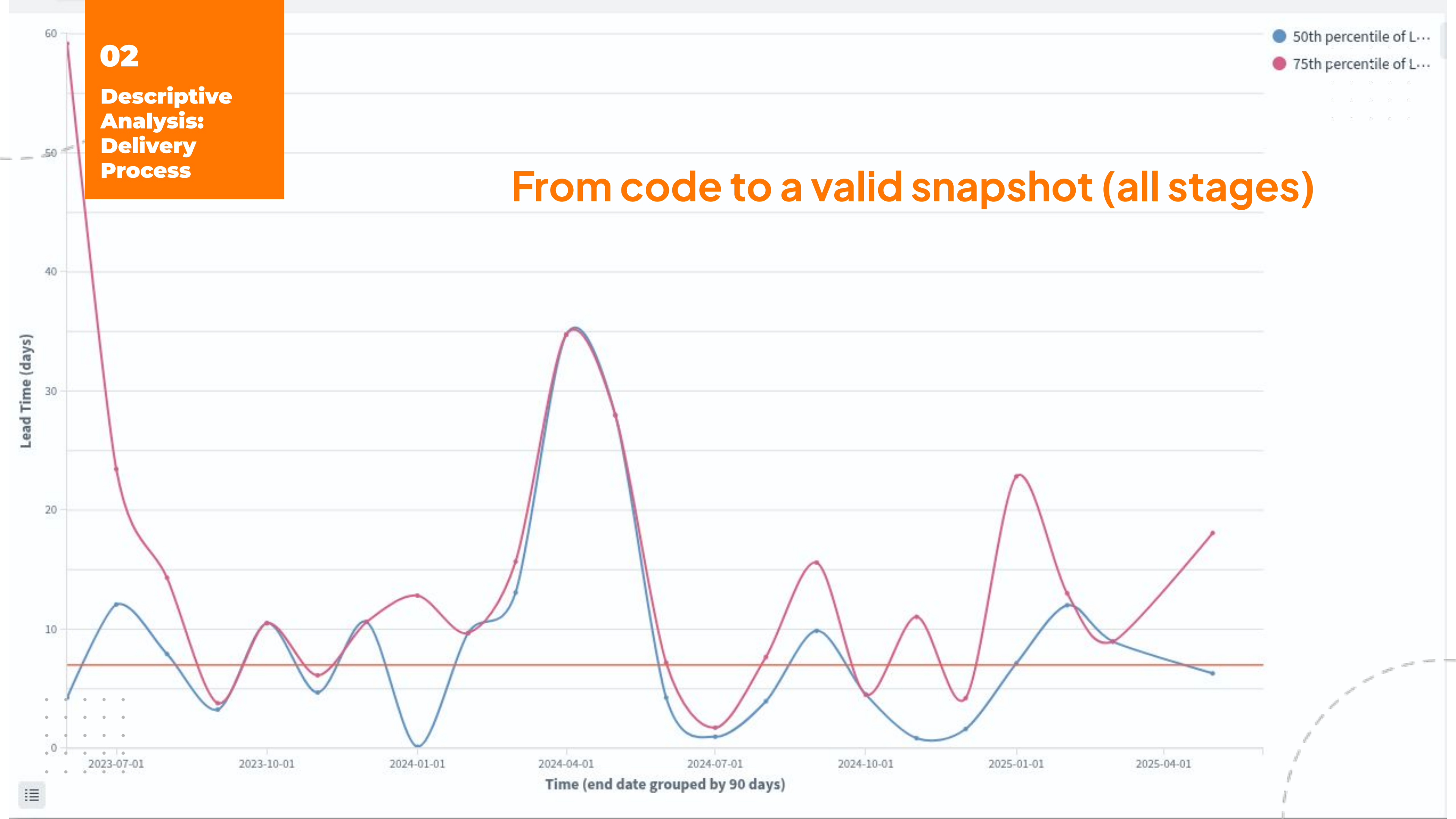
- Failure Recovery Time: when something goes wrong, the time your process takes to produce a successful output
- Change Failure Rate: successes versus failure



02
**Descriptive Analysis:
Delivery Process**

From code to a valid snapshot (all stages)

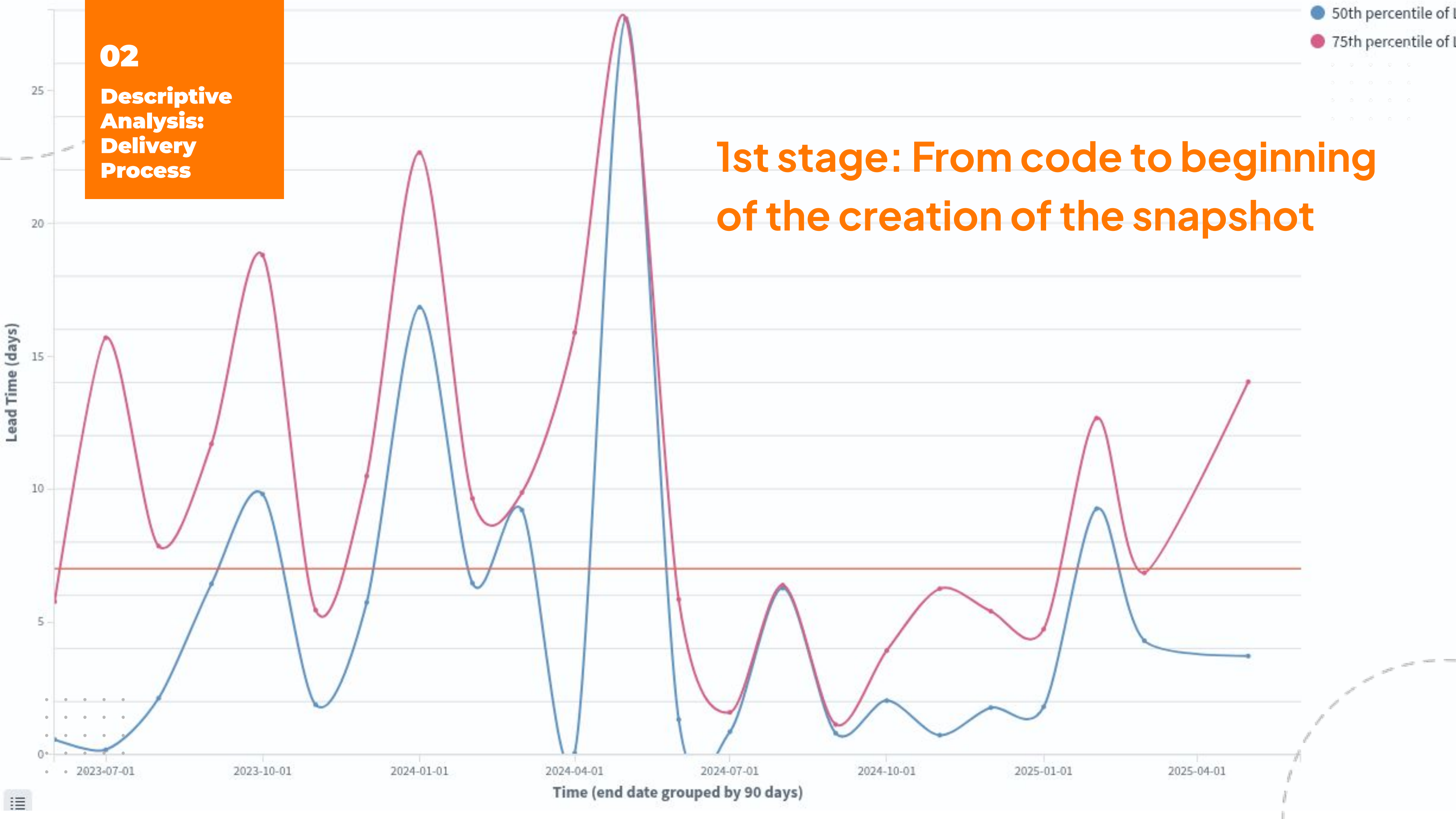
- 50th percentile of L...
- 75th percentile of L...



02

Descriptive Analysis: Delivery Process

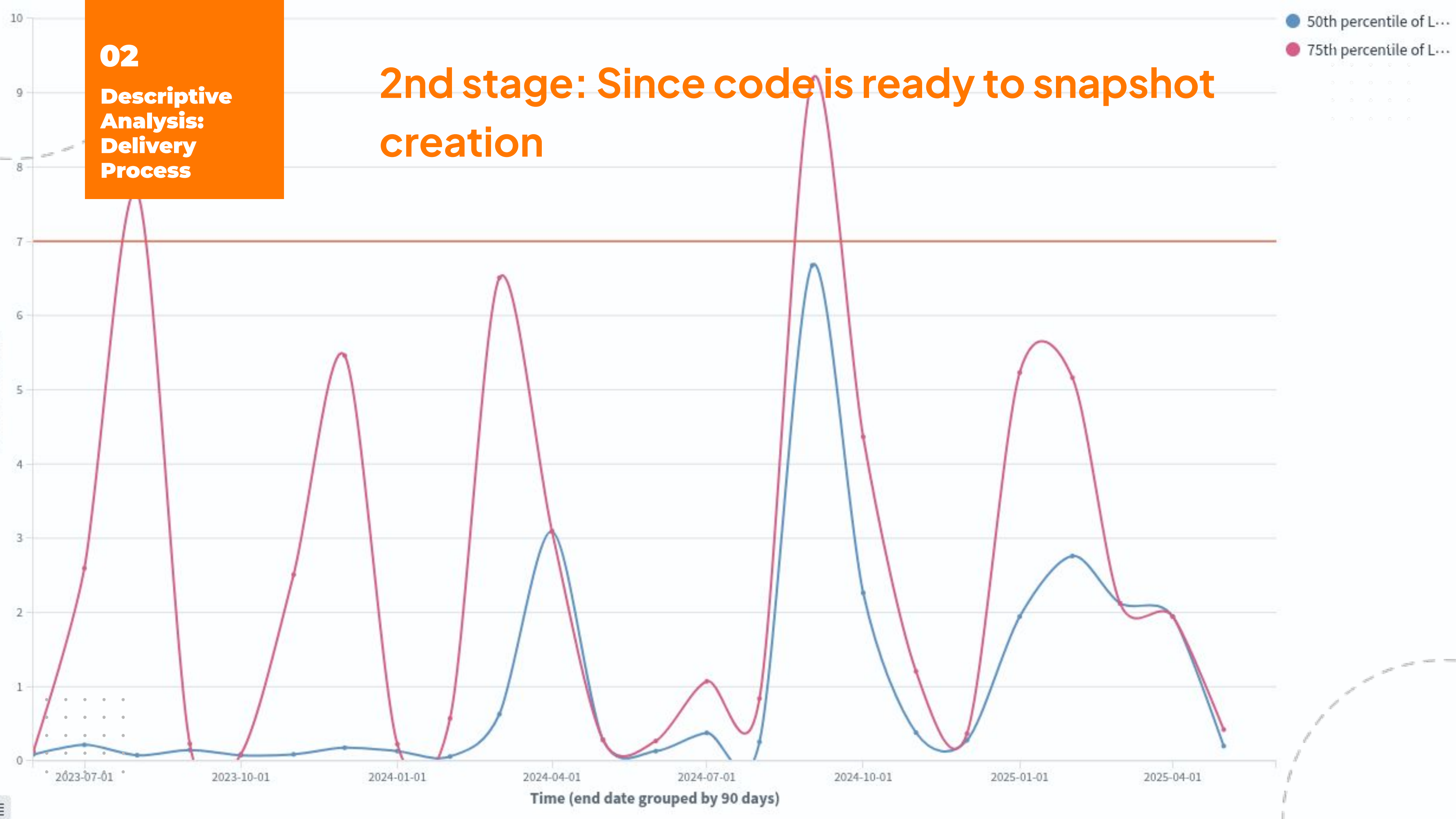
1st stage: From code to beginning
of the creation of the snapshot



02

Descriptive
Analysis:
Delivery
Process

2nd stage: Since code is ready to snapshot
creation



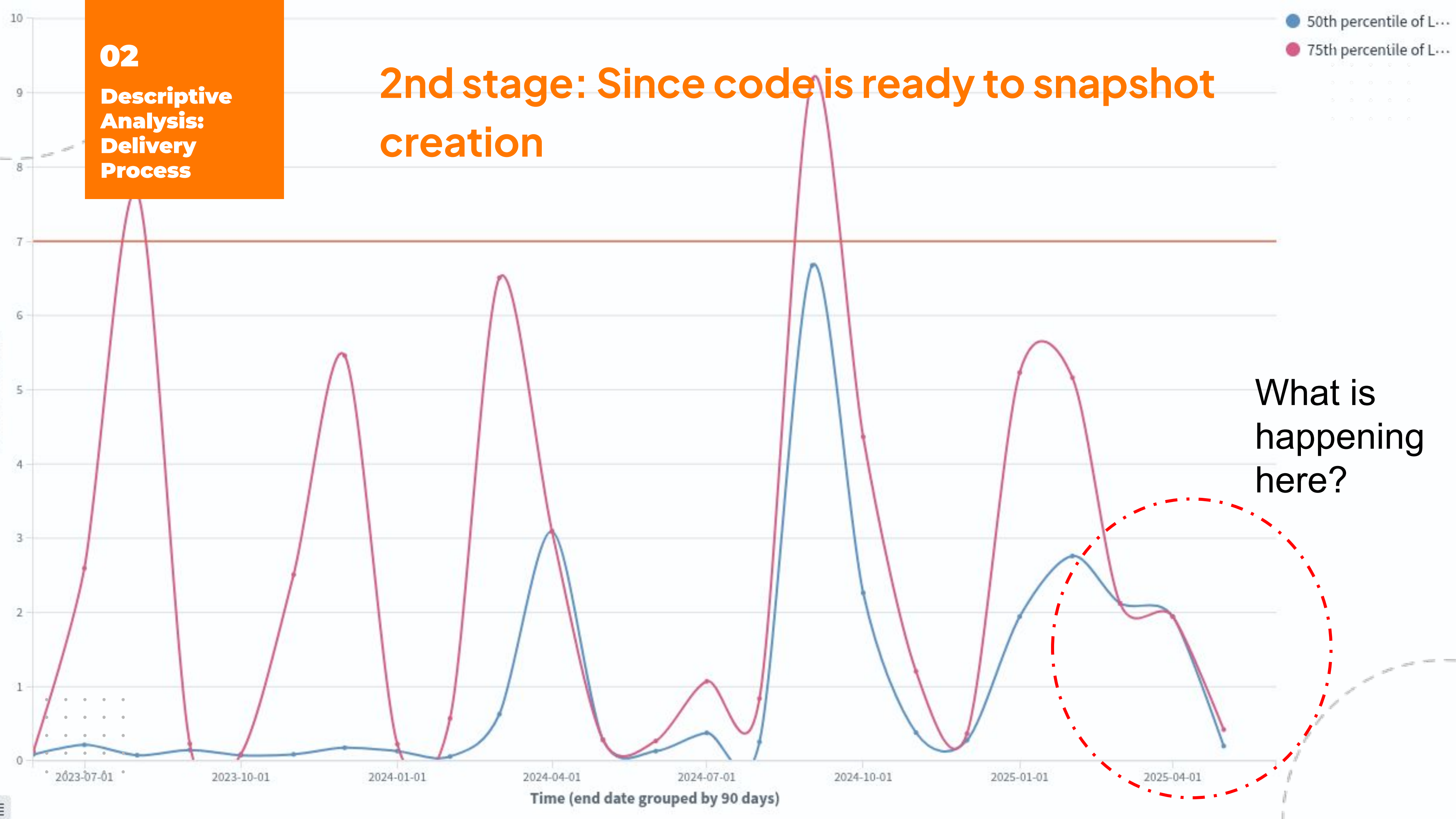
02

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50th percentile of L...

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What is
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here?

02

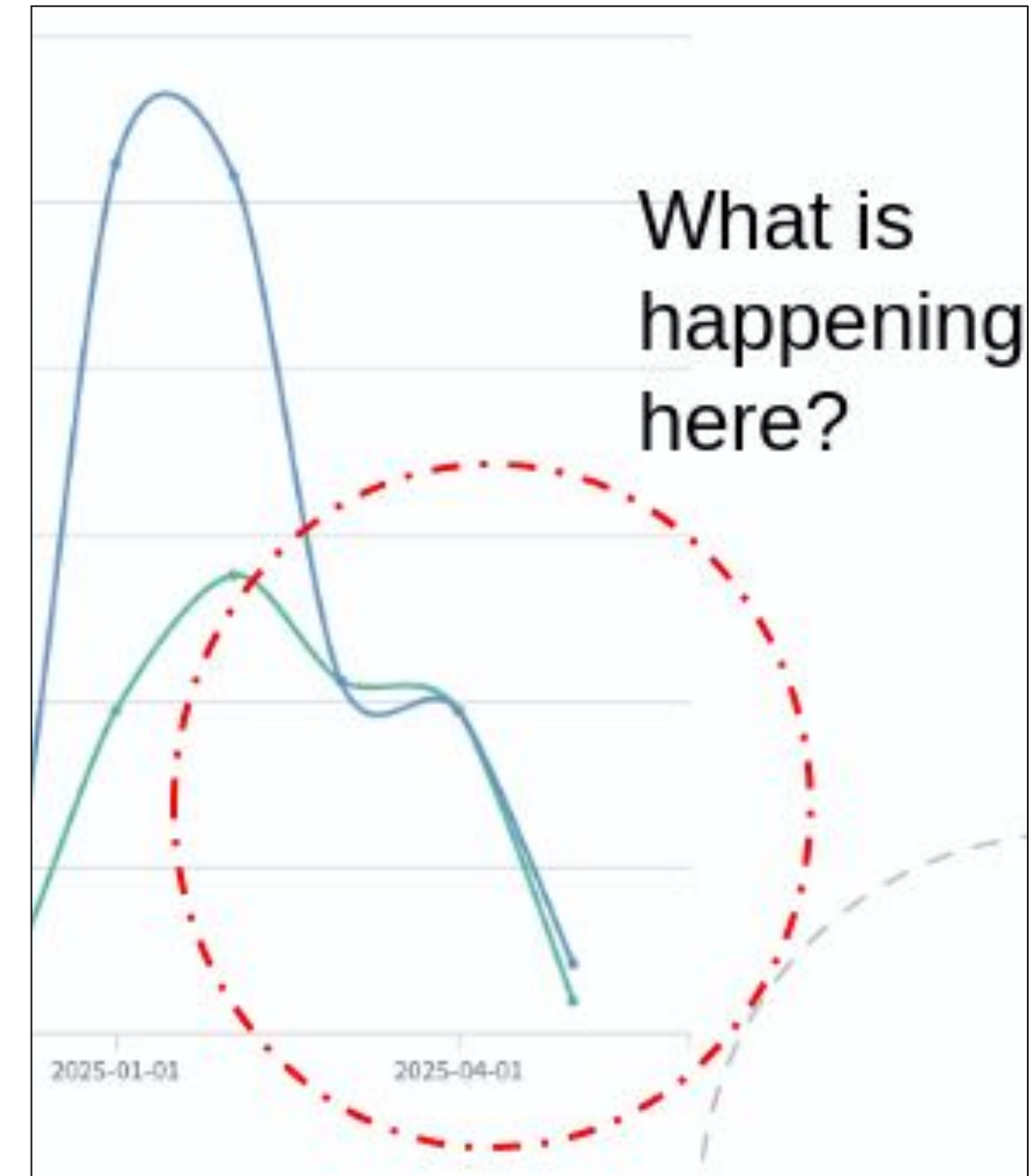
Descriptive Analysis: Delivery Process

What is happening here? Why such a big reduction?

Is this the result of an improvement?

Is this the result of a lack of activity?

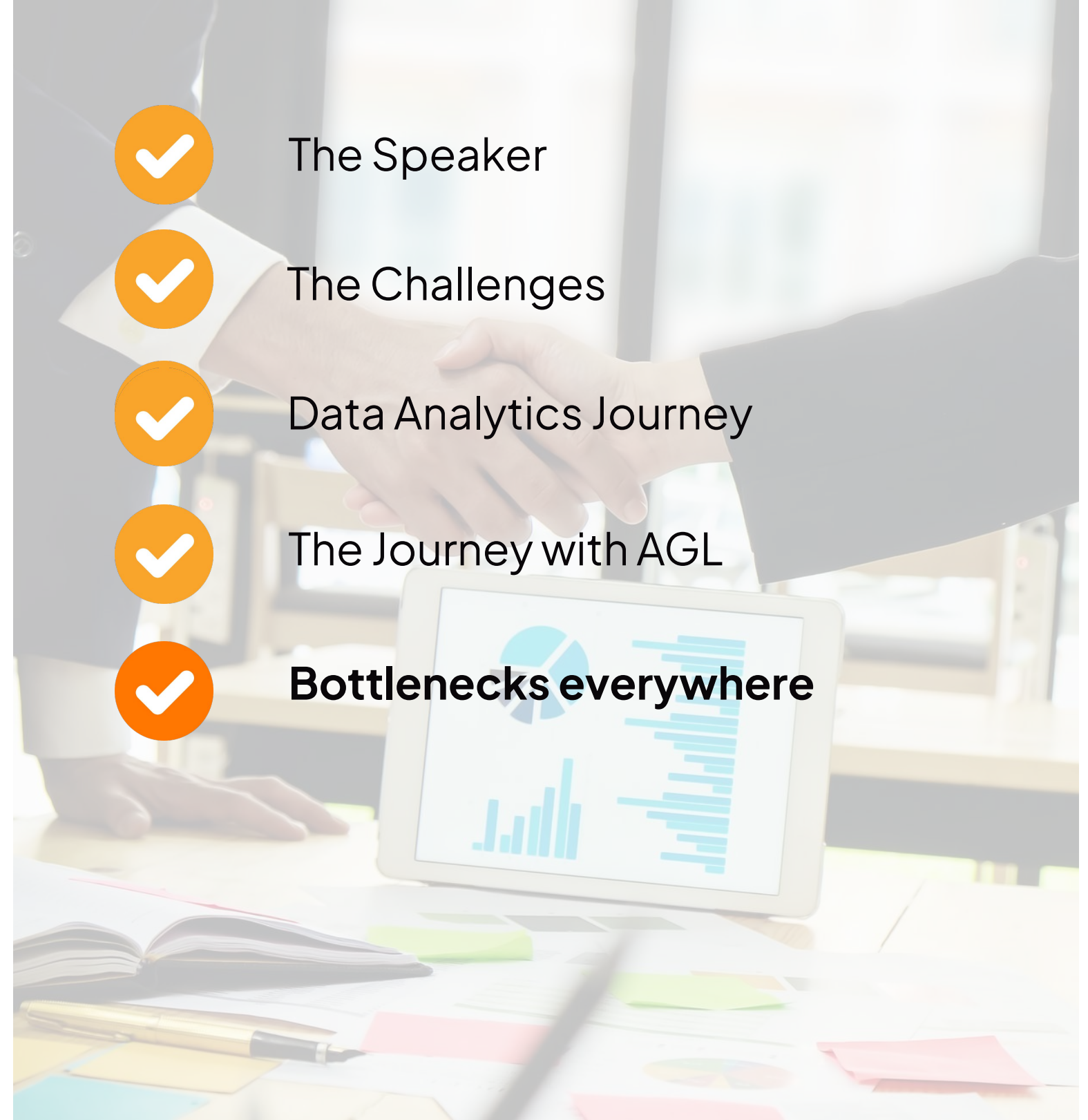
What was the effect of this in the overall picture?



Bottlenecks. Example of an improvement confirmed by the data



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- ✓ Data Analytics Journey
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- ✓ **Bottlenecks everywhere**





Bottlenecks.

**Example of
improvement
s confirmed
by the data**



Jan-Simon: “There should be a decline in build time, because the builds are using NFS now”

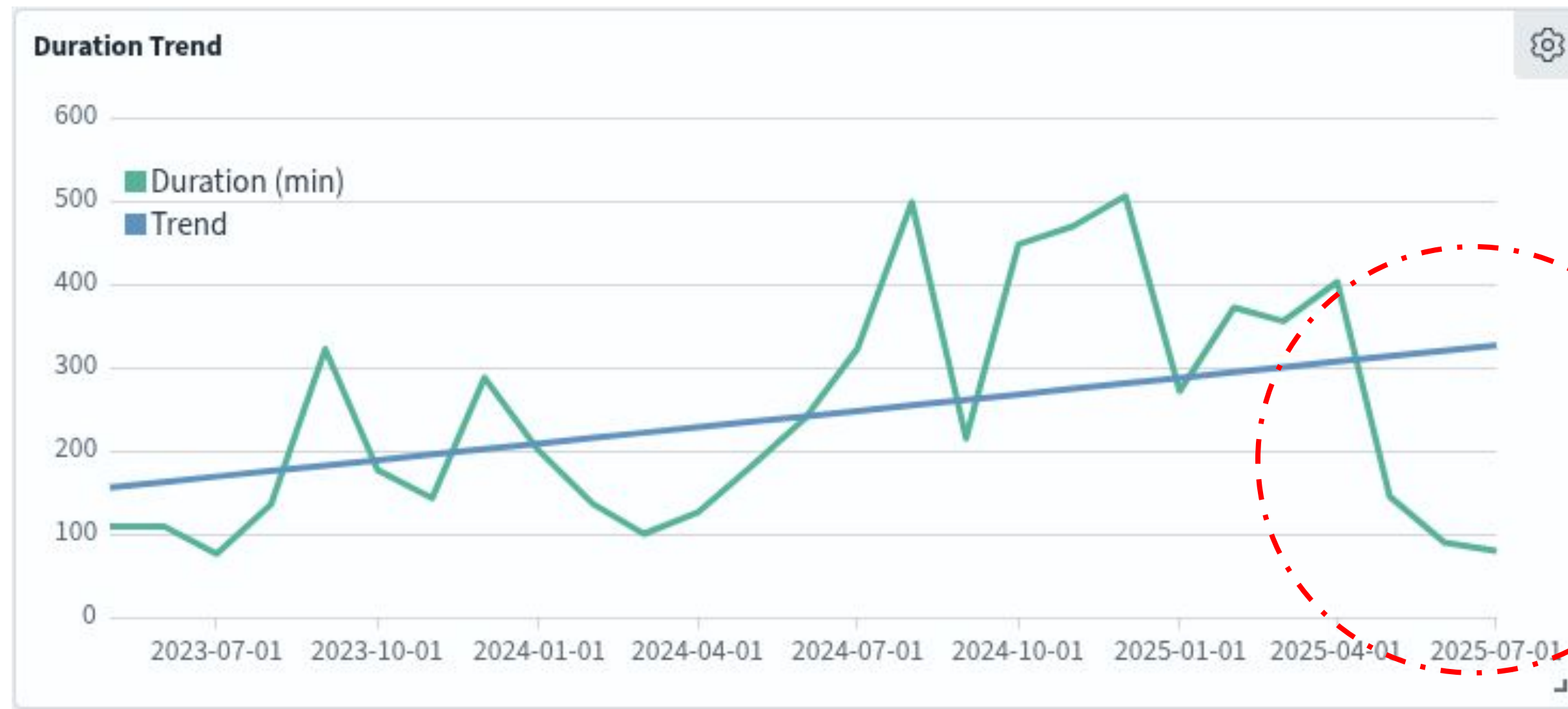
Let’s see what the data says ..



Bottlenecks.

**Example of
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by the data**

Data from the Jenkins job name ci-platform-AGL-repo-verify





Bottlenecks.

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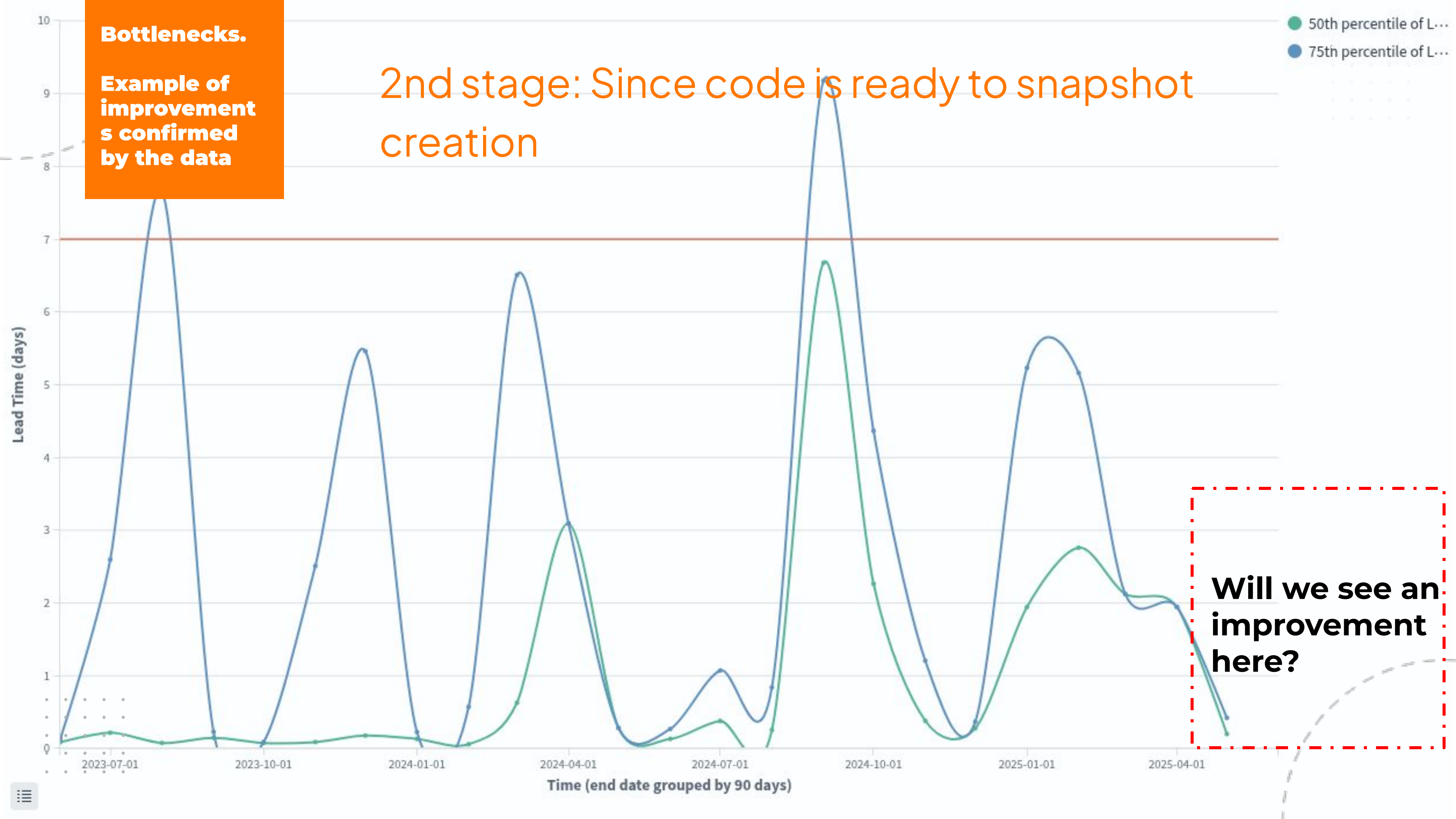
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Bottlenecks.

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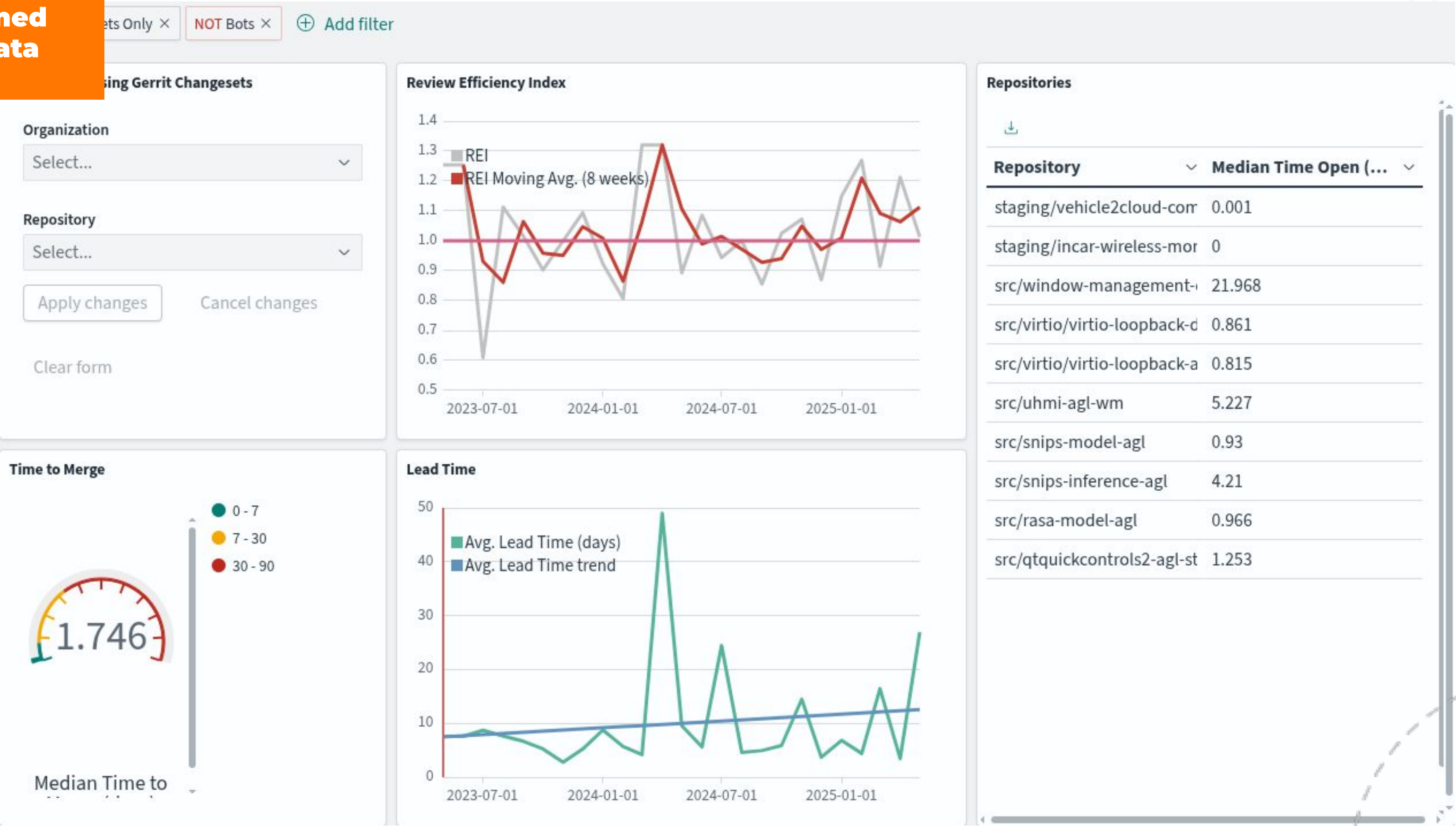
We've seen the commit stage lead time is high.

Can we identify relevant bottlenecks?



Bottlenecks.

Example of
improvement
s confirmed
by the data



References



Talks

- Improving Delivery Process Performance for SDV Products Helped by Data Analytics. [Video](#)
- Learn About AGL's Delivery Process Through Data And Analytics. [Video](#)
- [Code Reviews](#). DORA Community Discussions

Articles

- [Delivery Performance Analytics](#) service description, by Bitergia
- [Introducing Delivery Performance Analytics](#) service, by Agustin B.B.
- Improve your software product delivery process performance using metrics [\[1\]](#)[\[2\]](#)

Documentation

- AGL wiki: [AGL's Delivery Process Performance](#)
- Bitergia Analytics Platform ([BAP](#)) open source data analytics tool. [Code](#).
- [CHAOSS](#) project metrics (Bitergia is co-founder)

Delivery Performance Analytics,
a service from Bitergia and Toscalix
Consulting that applies business
intelligence to the production of
software-defined products, to improve
delivery and organizational performance as
well as staff's well-being



Documentation



Luis Cañas-Díaz

